

WARNING: THIS REVIEW IS NOT MEANT TO BE REFLECTIVE OF THE ACTUAL EXAM. THIS IS SIMPLY A SET OF SOME EXTRA PROBLEMS ON THE EXAM MATERIAL. IT IS NOT EXHAUSTIVE OF ALL TYPES OF MATERIAL OR QUESTIONS. YOU SHOULD USE THIS AS ONE STUDY TOOL, NOT YOUR ONLY STUDY TOOL!

1. Consider the following scenario:

Pollsters wanted to determine the proportion of registered voters in April 2019 who approved of President Trump's performance. They called 5000 randomly selected registered voters and asked 4123 of those (the rest weren't home, didn't answer, or hung up) "Do you approve of Pres. Trump's performance?"

- (a) What is the population of interest?
 - (b) How many people were in the sample?
 - (c) What is the variable of interest? Is it categorical (ordinal/nominal) or quantitative (discrete/continuous)?
 - (d) What is the parameter of interest? What statistics are they using to estimate it?
2. A store stocks lightbulbs from 3 suppliers (A, B, and C). Supplier A provides 80% of the store's lightbulbs, while suppliers B and C provide 15% and 5%, respectively. Of lightbulbs from supplier A, 0.1% are defective; of those from supplier B, 0.3% are defective; and of those from supplier C, 0.5% are defective.
- (a) **What is the probability of a customer buying a lightbulb that is from supplier A and is defective?**
 - (b) **What is the probability of a customer buying a defective bulb?**
 - (c) **Given the bulb is not defective, what is the probability it came from supplier C?**
3. There is 75% chance that event B will occur. If event B doesn't occur, then there is 15% chance that event C will occur. **If event C does not occur, compute the probability that event B does not occur. Assume $P(B \cap C) = 0.2$.**
4. (Fall 2019 Exam 1) A manufacturer buys 60% of a raw material from Supplier 1 (S1), 30% from Supplier 2 (S2) and 10% from Supplier 3 (S3). There are no other suppliers used by the manufacturer. Of those raw materials supplied by S1, 2% are defective. For S2 and S3, 4% and 3% of the raw materials are defective respectively. A piece of raw material is selected from the production line at random.
- (a) **What is the probability the raw material selected is defective? What probability rule are you using here?**
 - (b) **If the raw material selected was defective, what is the probability it came from S2?**
 - (c) **If the raw material selected was not defective, what is the probability it came from S3?**

5. (Spring 2019 Exam 1) In a recent report from the US Health Resources and Services Administration, the following population-level characteristics were presented:
- 2% of Americans were infected with HCV
 - among those Americans infected with HCV, 5% were infected with HIV
 - among those Americans infected with HIV, 12% were infected with HCV.
- (a) **Find the probability that an individual has HIV.**
- (b) **Are the two viral diseases independent? Prove your answer.**
6. Togolese scientists have identified a new strain of the H1N1 virus. The genetic sequence of the new strain consists of alterations in the hemagglutinin protein, making it significantly different than the usual H1N1 strain. Public health officials wish to study the population of residents in Lomé. Suppose that in this population,
- the probability of catching the usual strain is 0.10
 - the probability of catching the new strain is 0.05
 - the probability of catching both strains is 0.01.
- (a) **Find the probability of catching at least one of the strains of H1N1 virus. What probability rule is used here?**
- (b) **Find the probability of catching the usual strain, given that the new strain is caught.**
- (c) **Find the probability of not catching the usual strain, given that the new strain is not caught.**
7. Go over the homework.